Lab 7
Advanced Combinational Logic

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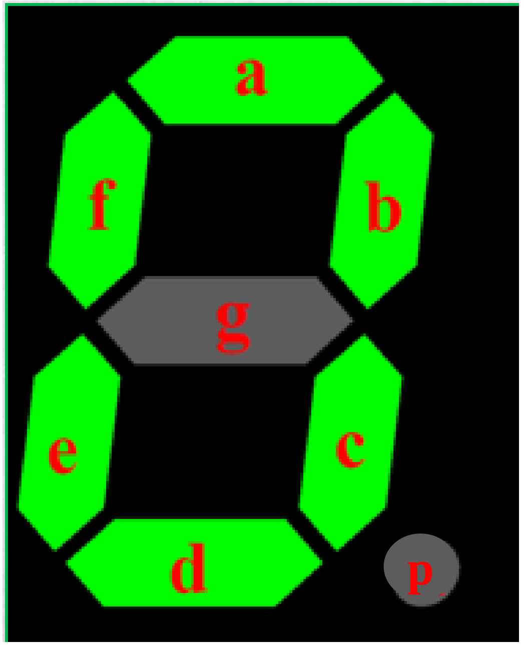
Contents of this slide are provided by NTOU VLSI Lab601

Objectives

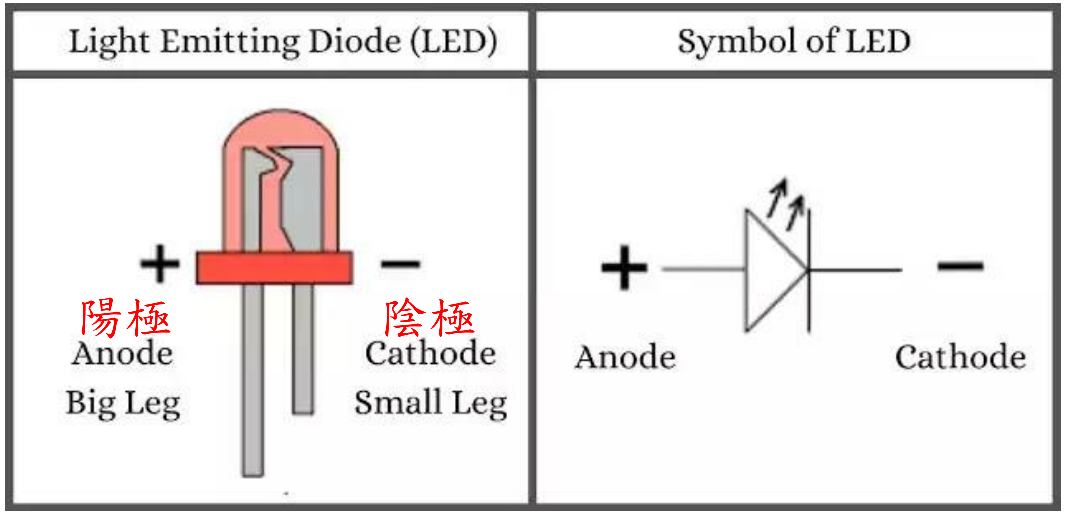
- Applications of seven-segment display (SSD) (七段顯示器)

Structure of An SSD

- An SSD consists of 7 segments and a dot.



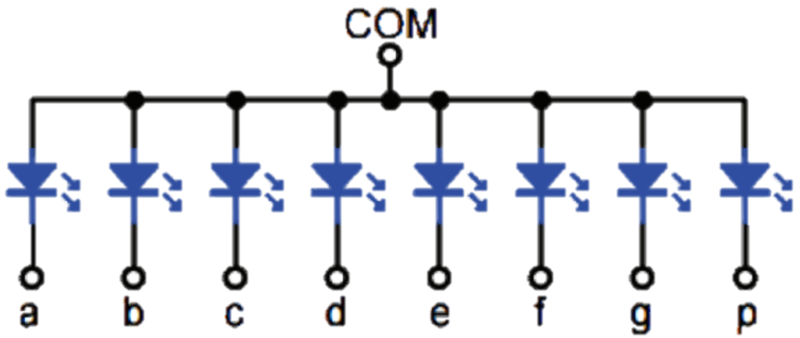
- Each segment is realized by an LED.



Source: https://www.atlearner.com/2020/07/what-is-an-led.html

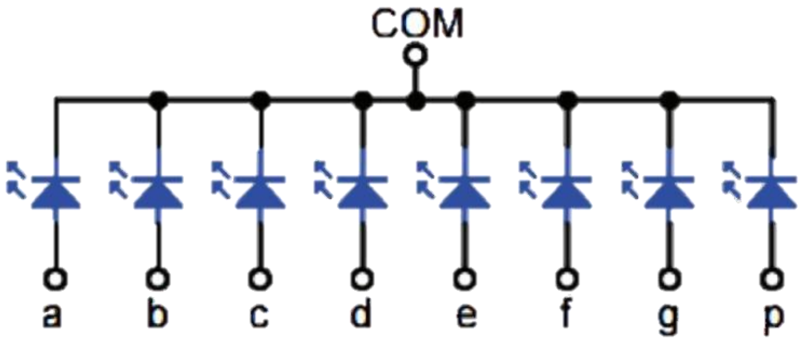
Types of SSD – Common Anode (共陽極)

- All the anodes of the LEDs are connected together.
- How to operate?
  - Always connect the COM pin to a logic HIGH or 5 V.
  - Connect the segment(s) that need to be lit up to a logic LOW or GND.

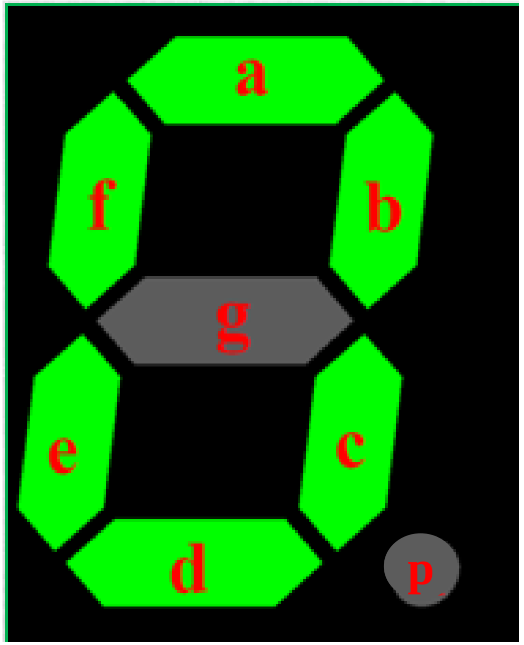


Types of SSD – Common Cathode (共陰極)

- All the cathodes of the LEDs are connected together.
- How to operate?
  - Always connect the COM pin to a logic LOW or GND.
  - Connect the segment(s) that need to be lit up to a logic HIGH or 5 V.

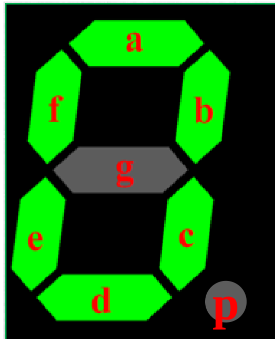


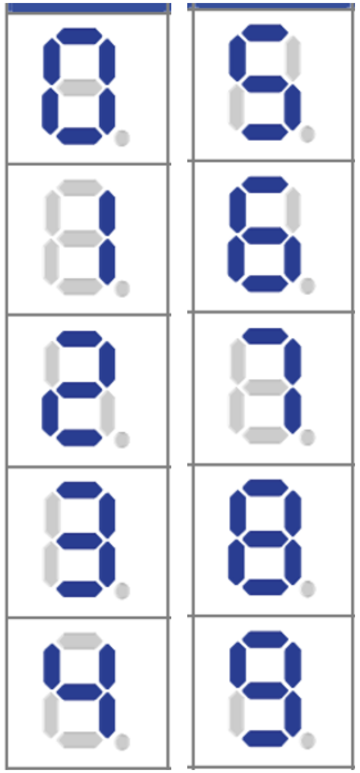
Control The SSD



MSB							LSB
p	g	f	e	d	c	b	a

Truth Table for Common Anode SSD

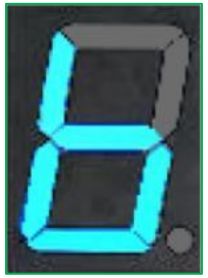




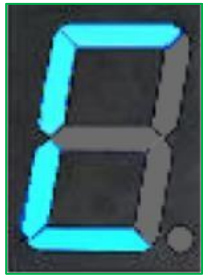
Input					Output						
a_3	a_2	a_1	a_0		g	f	e	d	c	b	a
0	0	0	0	0	1	0	0	0	0	0	0
0	0	0	1	1	1	1	1	1	0	0	1
0	0	1	0	2	0	1	0	0	1	0	0
0	0	1	1	3	0	1	1	0	0	0	0
0	1	0	0	4	0	0	1	1	0	0	1
0	1	0	1	5	0	0	1	0	0	1	0
0	1	1	0	6	0	0	0	0	0	1	0
0	1	1	1	7	1	1	1	1	0	0	0
1	0	0	0	8	0	0	0	0	0	0	0
1	0	0	1	9	0	0	1	0	0	0	0
1	0	1	0	A	0	0	0	1	0	0	0
1	0	1	1	b	0	0	0	0	0	1	1
1	1	0	0	C	1	0	0	0	1	1	0
1	1	0	1	d	0	1	0	0	0	0	1
1	1	1	0	E	0	0	0	0	1	1	0
1	1	1	1	F	0	0	0	1	1	1	0

Examples

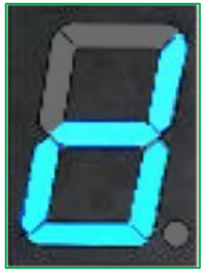
A (10)
1000-1000



b (11)
1000-0011



C (12)
1100-0110



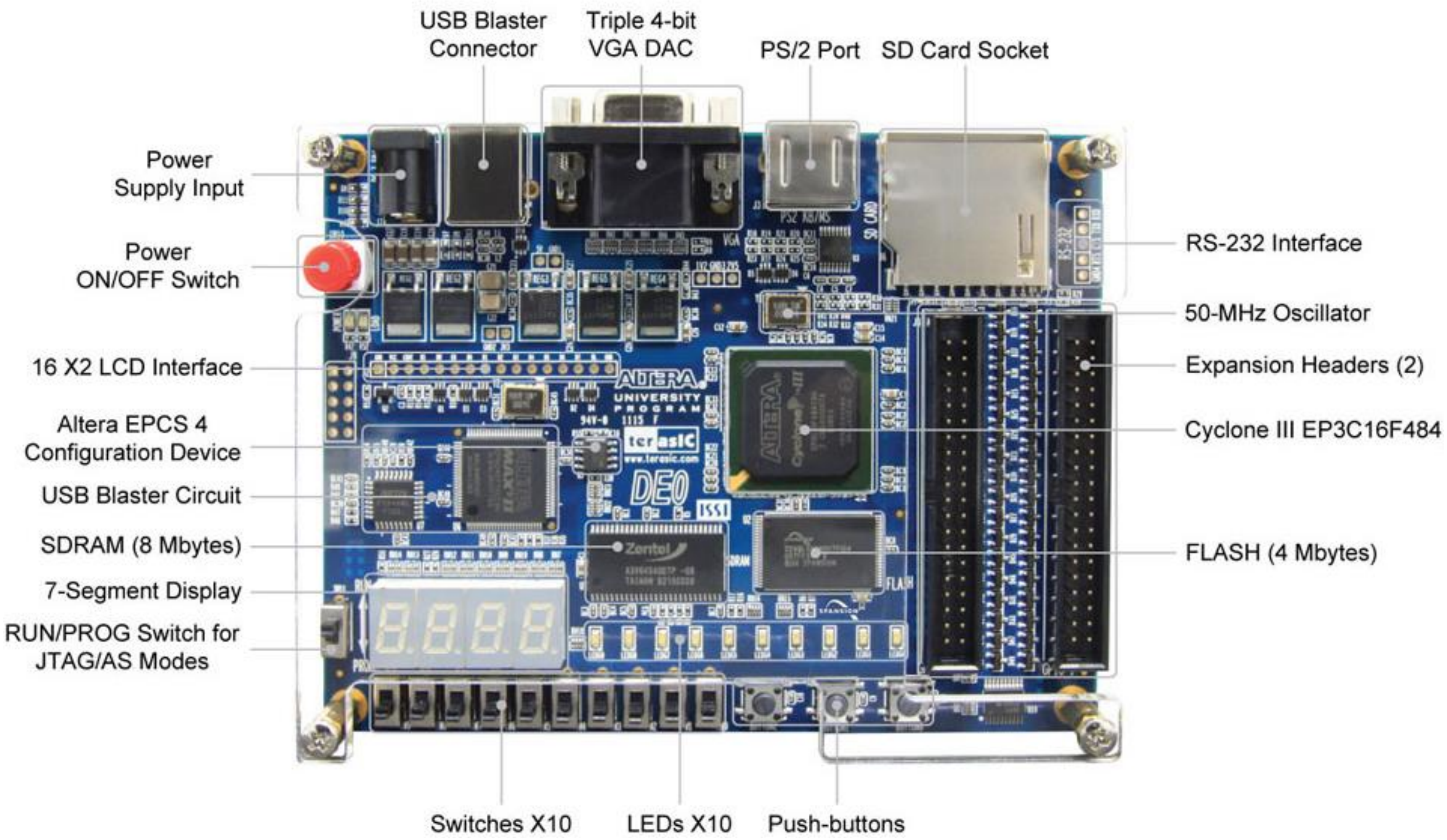
d (13)
1010-0001

E (14)
1000-0110

F (15)
1000-1110

Development and Education Board – DE0

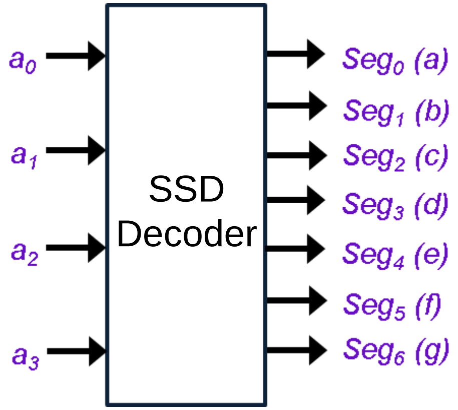
- The online resources of DE0 can be downloaded from here



LAB 1 – SSD Decoder (1/2)

- Program an SSD decoder, perform signal simulation, and download to the DE0 board
- Signal and pin assignments:

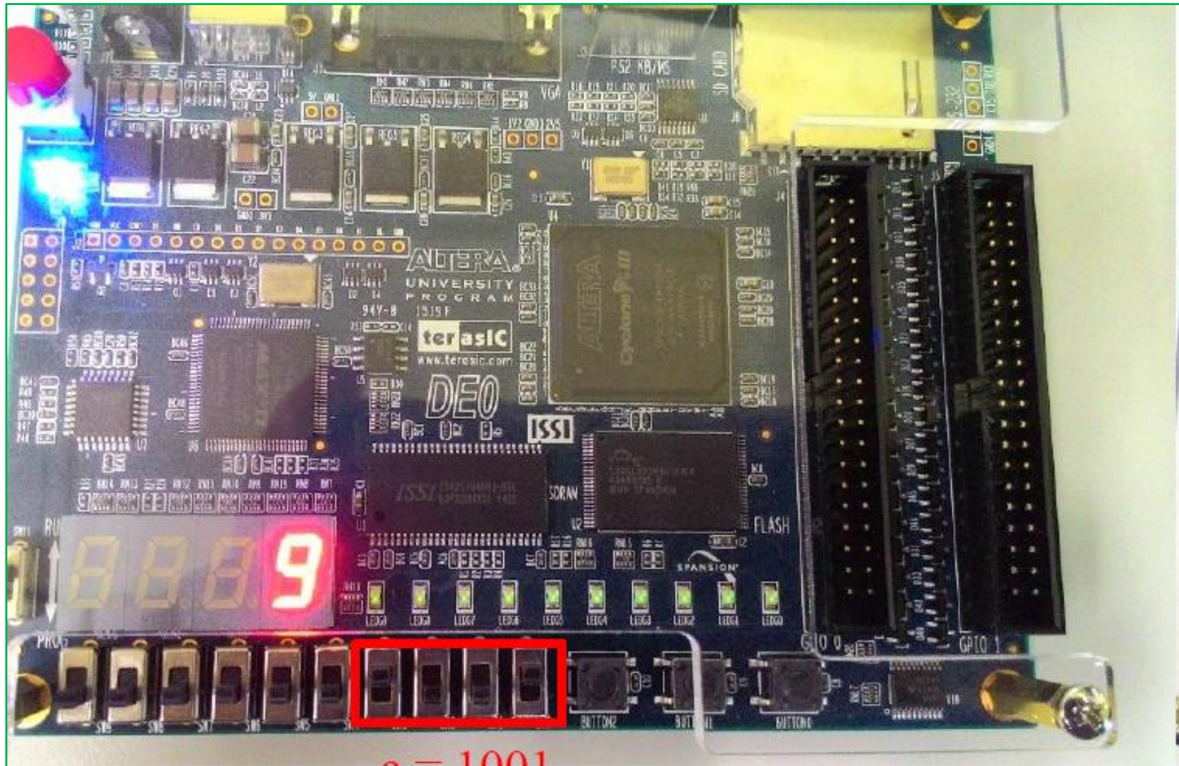
Signal	Type	DE0 pin
a_0 - a_3	Input	Switch (SW[0]-SW[3])
Seg_0 (a) - Seg_7 (p)	Output	SSD (HEX0)



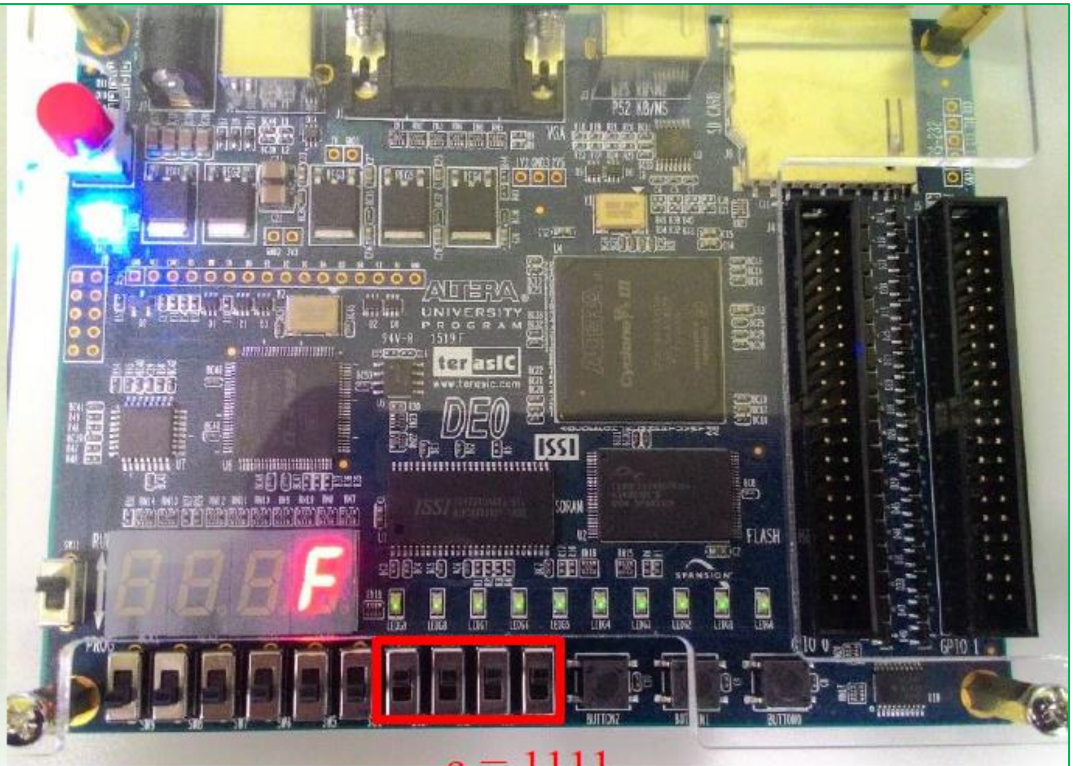
When downloading to the DE0, only the above pins can be used.

- Testbench: The SSD shows all the values from 0 to F

LAB 1 – SSD Decoder (2/2)



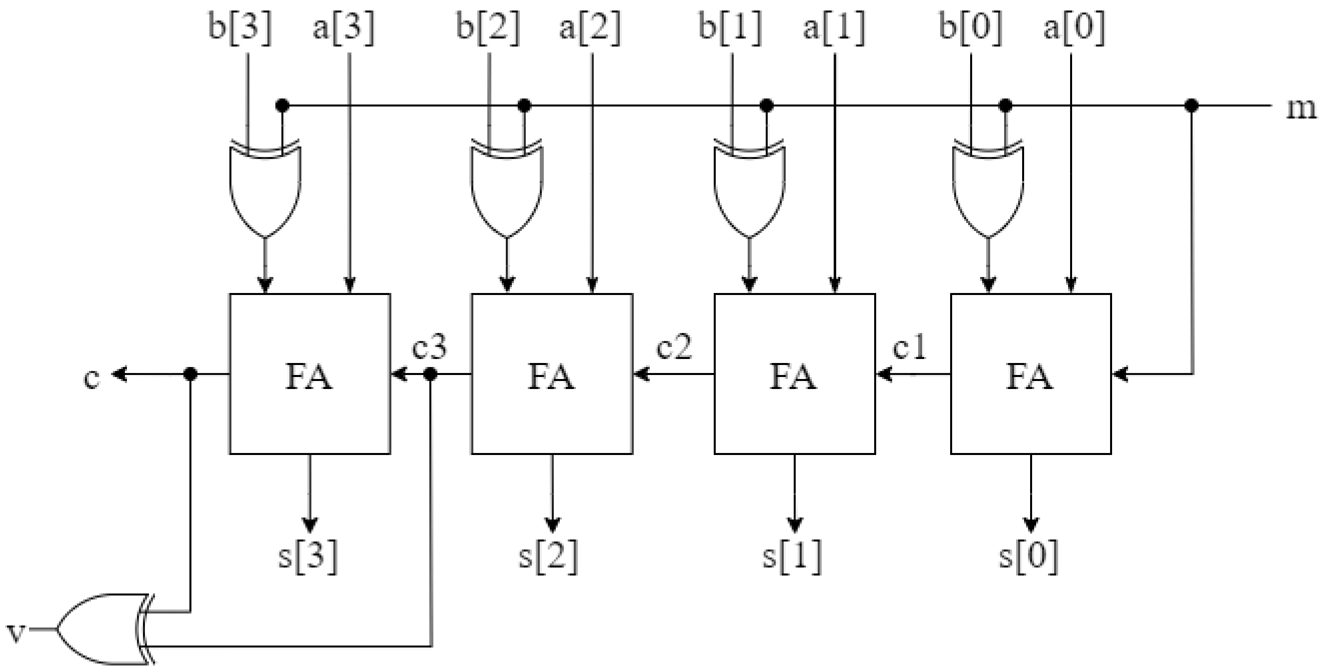
a = 1001



a = 1111

LAB 2 – 4-bit Adder/Subtractor

- Run ModelSim Simulation
- Signal assignments:
  - Input: a[3:0], b[3:0], m (perform addition (m=0) or subtraction (m=1))
  - Output: s[3:0], c, v (if there is an overflow, v=1)

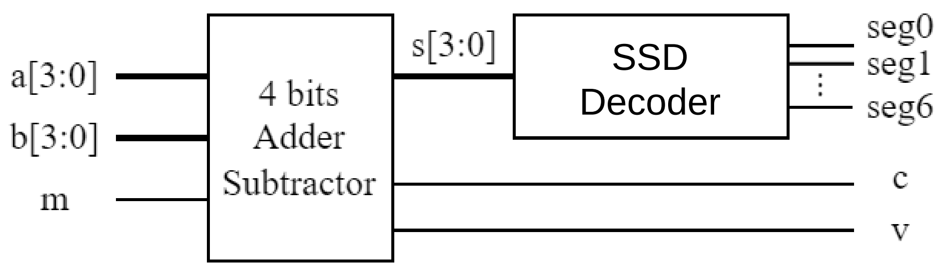


Testbench		
m	a[3:0]	b[3:0]
0	1111	1111
0	0100	0111
1	0000	1111
1	1101	1010
1	0111	1000

LAB 3 – 4-bit Adder/Subtractor + SSD

- Connect the 4-bit adder/subtractor with the SSD decoder, perform waveform simulation, and download to DE0 board.
- Signal and pin assignment:

Signal	Type	DE0 pin
a[3:0]	Input	Switch (SW[0]-SW[3])
b[3:0]		Switch (SW[4]-SW[7])
m		Switch (SW[8])
Seg_0 (a)-Seg_7 (p)	Output	SSD (HEX0)
c		LED (LED[0])
v		LED (LED[1])



- Testbench: The same as Lab 2.

Bonus – BCD Adder

- Implement a BCD adder, connect it to the SSD decoder, and download to DE0.
- Signal and pin assignments:

Signal	Type	DE0 pin
addend[3:0]		Switch (SW[0]-SW[3])
augend[3:0]	Input	Switch (SW[4]-SW[7])
carry_in		Switch (SW[8])
Seg_0 (a) - Seg_7 (h)		SSD (HEX0)
output_carry	Output	LED (LED[0])
carry_out		LED (LED[1])

  - output_carry ⇒ Decimal carry line
  - carry_out ⇒ Hexadecimal carry line
  - The second input of the second adder can be written as {1'b0, {2{output_carry}}, 1'b0}